

AAA Member Service Prediction and Travel Segmentation

Grant Robinson, Erik Herb, Jackson Swallow, Simon Salaj

Final project presentation



Business Question and Prediction Target

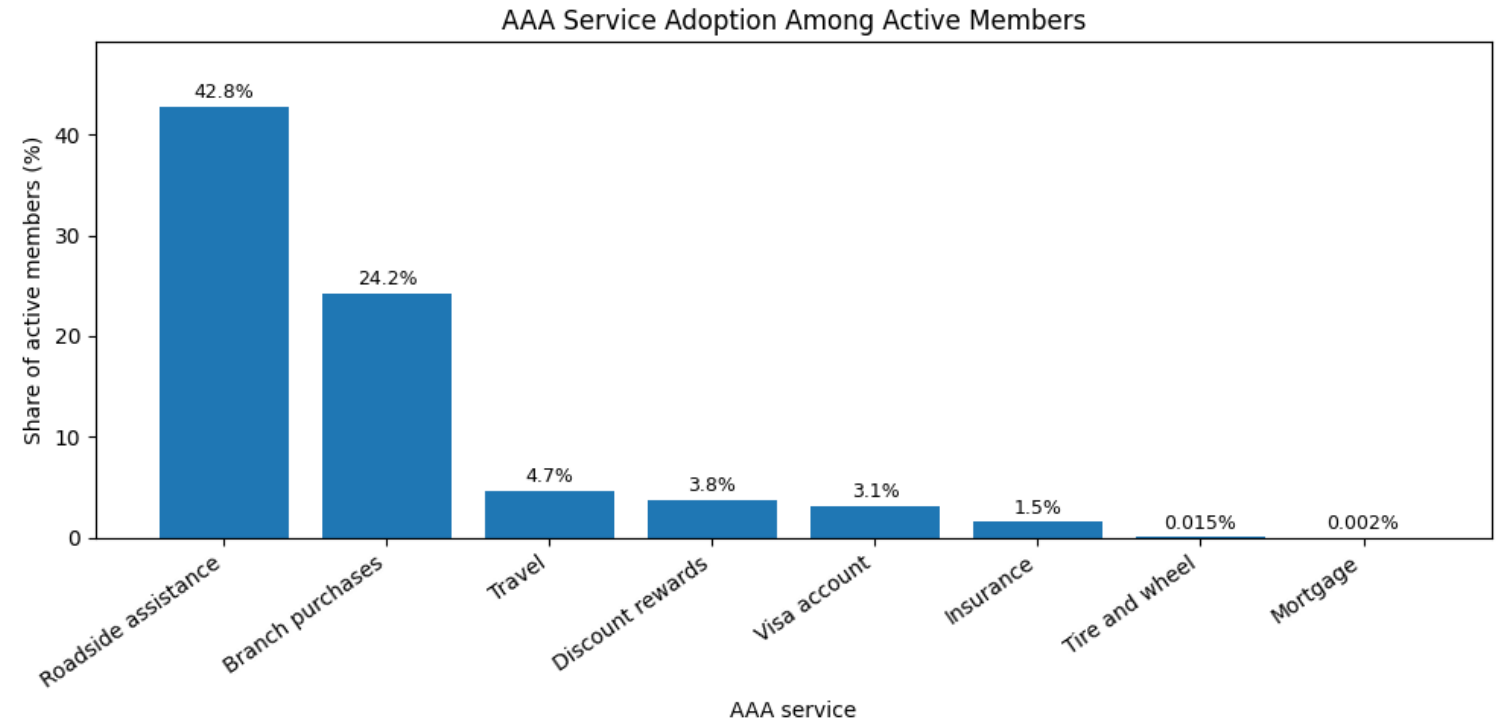
- Question: can we predict if a AAA member used roadside assistance?
- Target variable: roadside assistance
- Base population: 1,687,045 active AAA members
- Each row represents one active member after aggregation
- Positive class (label 1): 721,588 members used roadside assistance
- Negative class (label 0): 965,457 members did not use roadside assistance

Data Cleaning and Prep

- Aggregated to one row per member
- Selected active members only
- Lowercased and trimmed white spaces for all strings
- Set missing numericals to 0 and missing categorical to "na"
- Dropped exact duplicate rows from every table
- Converted date strings to date types
- Left class imbalance as is because:
 - Imbalance is mild (~44% / 56% split)
 - Our models (Random Forest and Logistic Regression) handle mild imbalances well
 - Minority class is still large (over 700k samples)
 - Rebalancing would distort real distribution

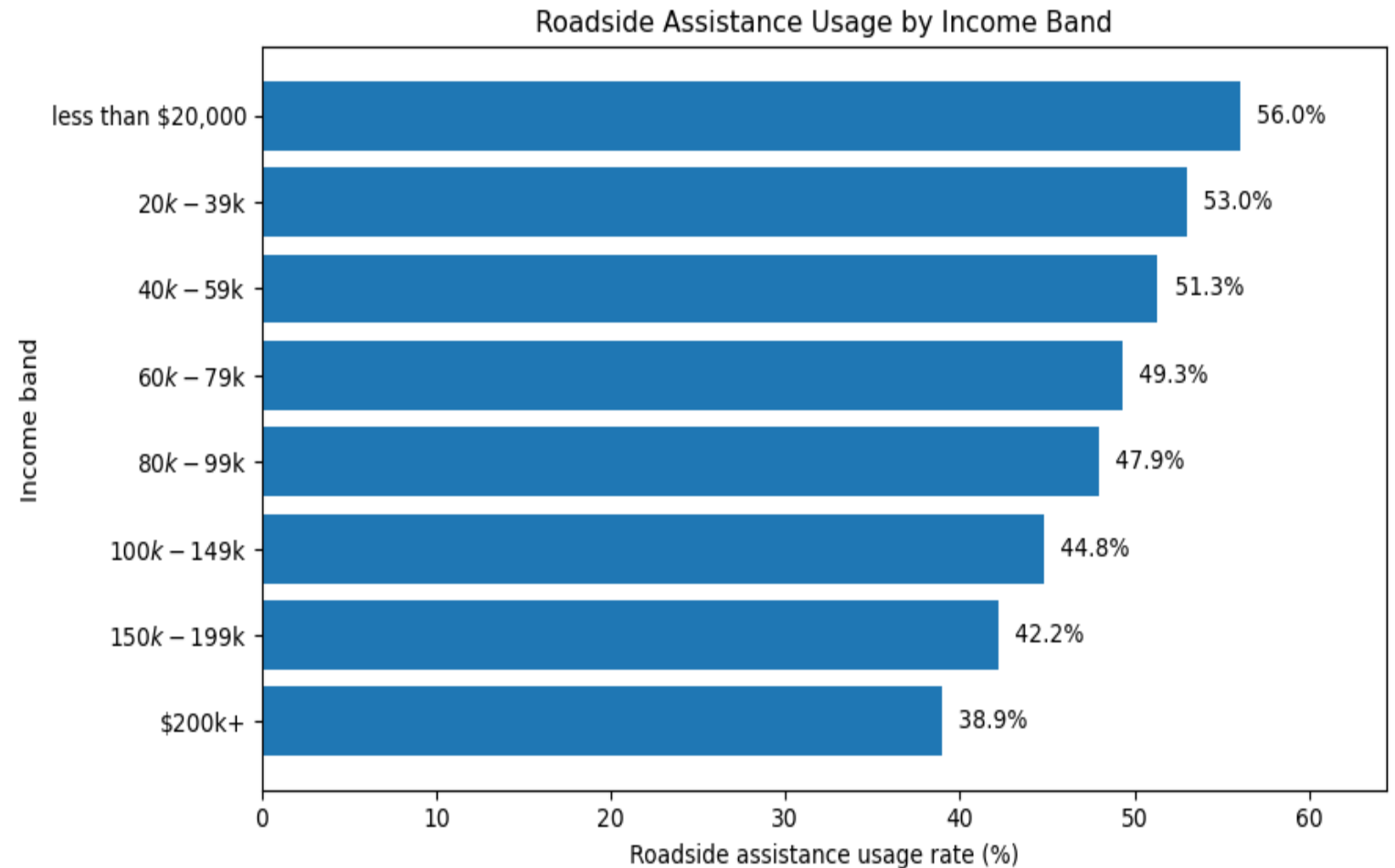
Exploratory Analysis: Service Adoption

- Roadside assistance is the largest non-transaction service behavior in the active-member table
- Branch purchases are common enough to provide useful activity signals
- Travel, Visa, insurance, tire/wheel, and mortgage activity are less common but still useful for segmentation



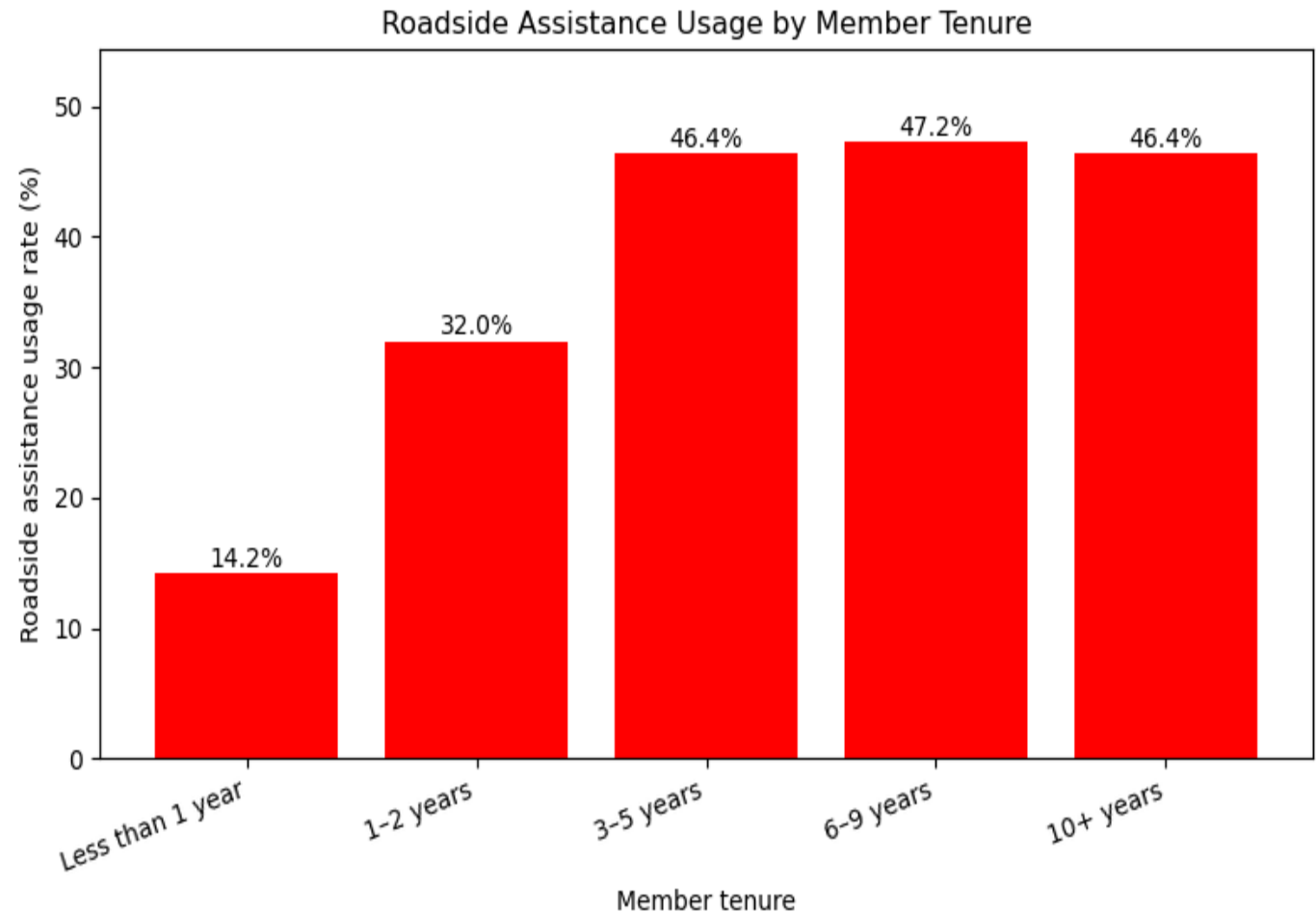
Exploratory Analysis: Roadside Usage Pattern

- Roadside assistance usage is highest among lower income bands
- Usage gradually declines as income bands increase
- This pattern supports using demographic and membership variables in the model
- The pattern provides a simple bridge from EDA into classification modeling



Exploratory Analysis: Roadside Usage by Tenure

- New members rarely use roadside assistance likely because they have not been a customer as long
- Usage rate rises steeply in the first 1–3 years, then flattens
- Suggests that the tenure of a customer is strong predictor for whether a customer has used roadside assistance



Modeling Approach

- Built a Spark ML pipeline with categorical and numeric features
- Used StringIndexer and OneHotEncoder for categorical variables in Logistic Regression
- Used VectorAssembler to create the final feature vector
- Created an 80/20 train/test split
- Class imbalance toward those who did not use roadside assistance
- Compared Logistic Regression, Random Forest, and Tuned Random Forest

Feature Groups Used

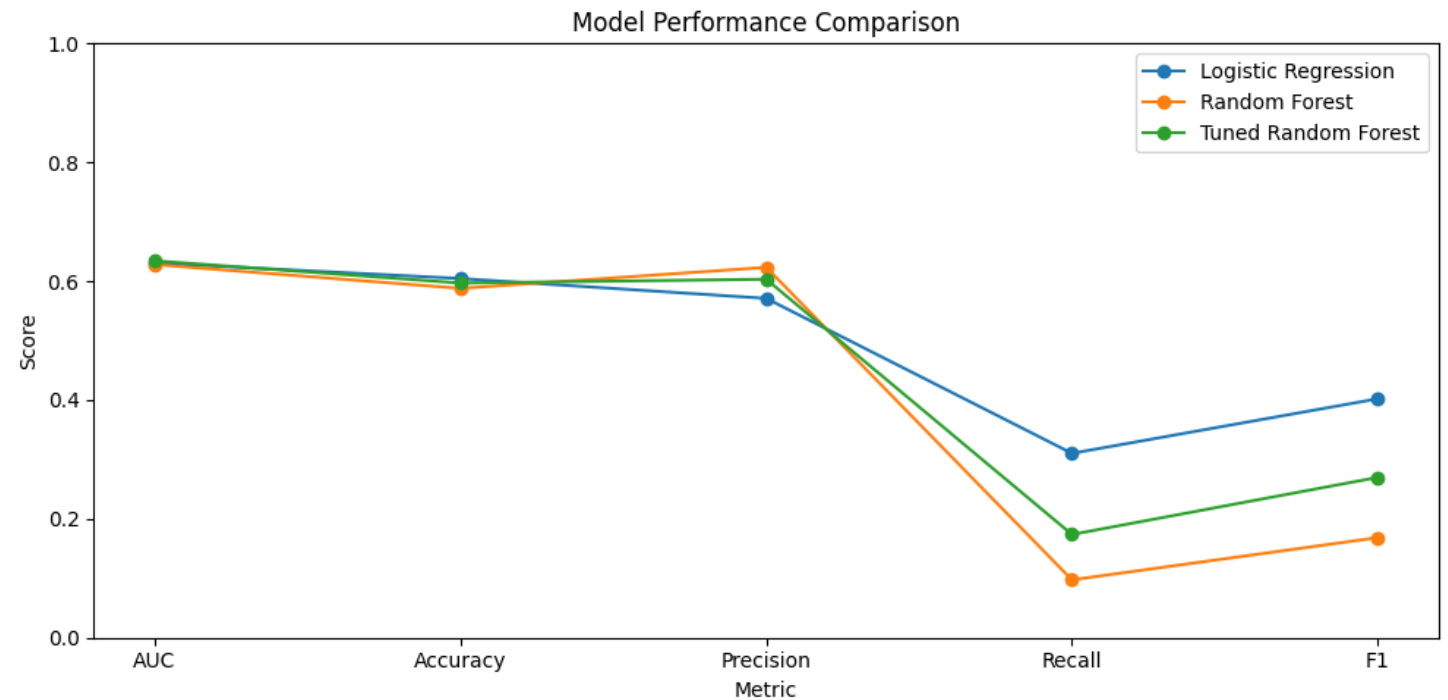
- Membership profile: tenure, category, coverage level, renewal method, and region
- Demographics: age group, income band, gender, education, marital status, and home ownership
- Service behavior: branch, discounts, roadside assistance, insurance, travel, Visa, mortgage, and tire/wheel activity
- Aggregated activity: usage flags, counts, totals, averages, and recency measures
- Member transaction activity was explored separately, but not used in the final model

Model Evaluation Metrics

- Evaluated models using:
- Area Under the Curve (how well a model can distinguish between different classes)
- Accuracy (how close model predictions are to the true, real-world value)
- Precision (how well the model can predict the positive true class)
- Recall (how many of all positive true class entries the model actually predicted)
- F1 score (measures a classification model's accuracy by balancing both precision and recall)
- confusion matrix counts (compares a model's predicted categories against the actual, true values)

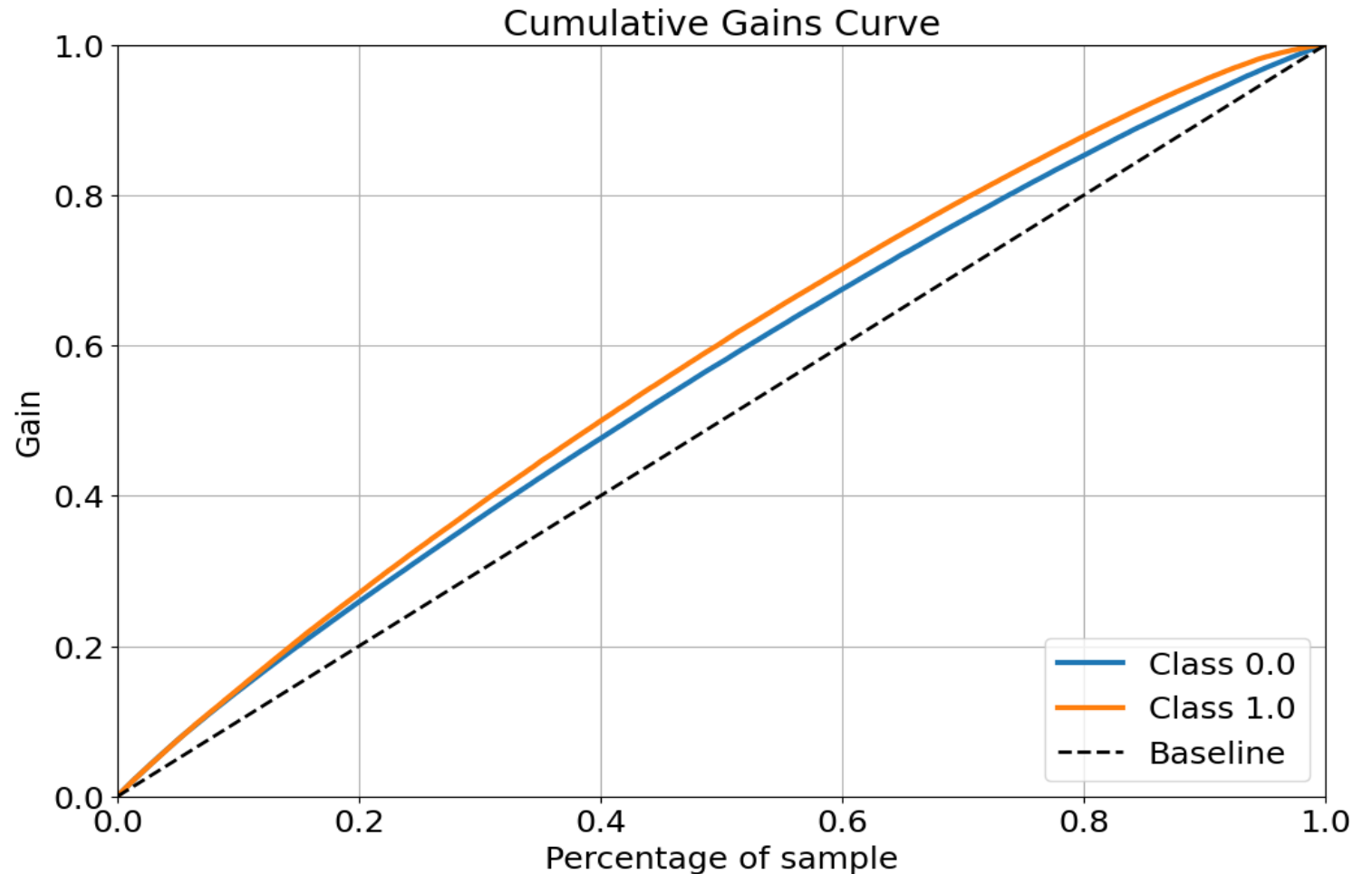
Model Performance

- Logistic Regression had the strongest area under the curve in this run
- Tuning improved Random Forest compared with the untuned version
- Recall remains the biggest area for improvement
- The models provide a baseline for future AAA service prediction work



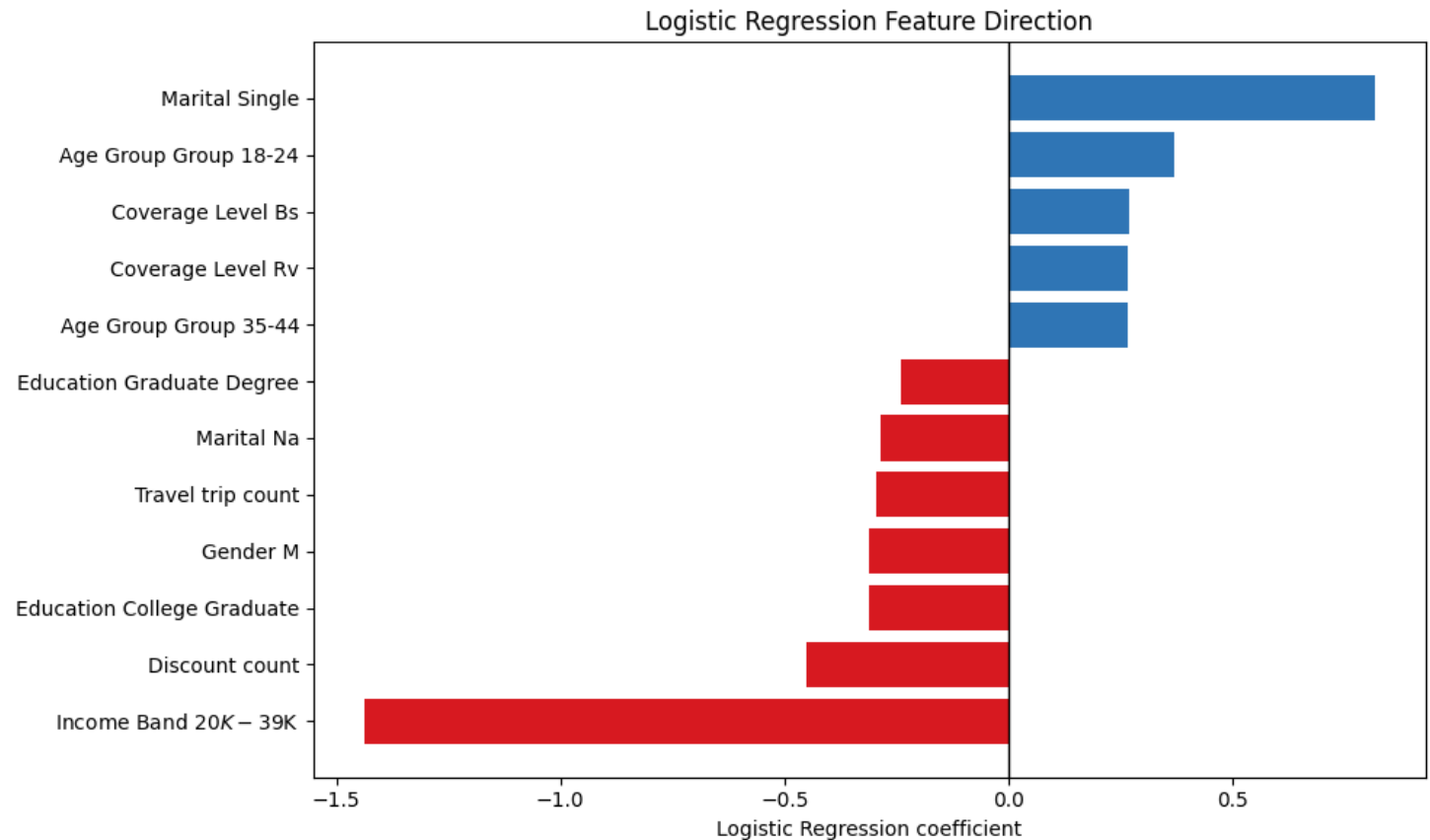
Logistic Regression Cumulative Gains Curve

- The model beats random guessing for both classes
- The members the model ranks highest tend to be the ones who use roadside
- The model's advantage is consistent across the entire curve rather than favoring a few high-ranking members
- Model can make outreach more efficient by focusing on higher-probability members



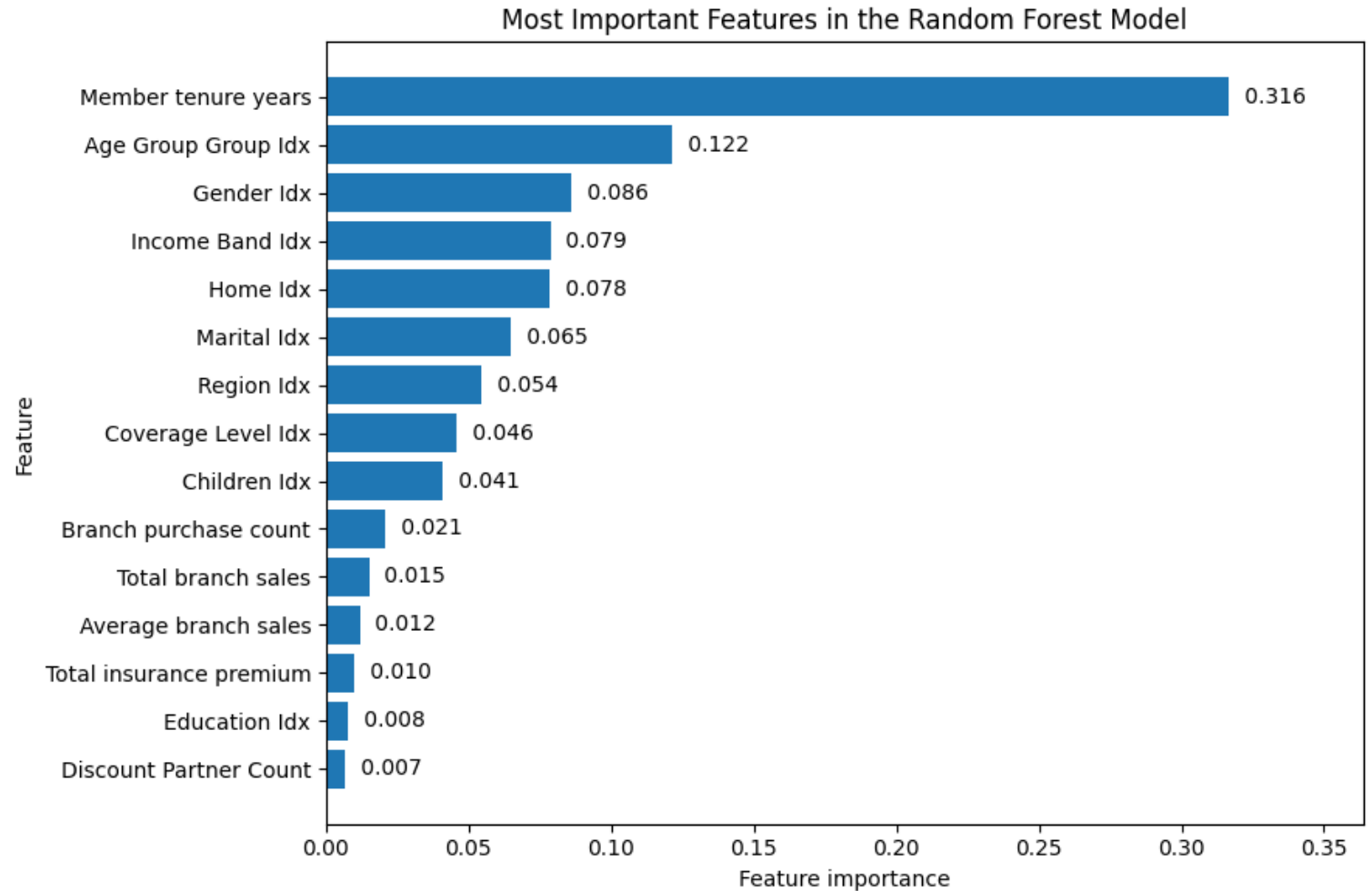
Logistic Regression Coefficient Direction

- Positive coefficients increase the predicted likelihood of roadside assistance usage
- Negative coefficients decrease the predicted likelihood
- This view shows direction, not only strength
- Demographic and service variables both contribute to the prediction



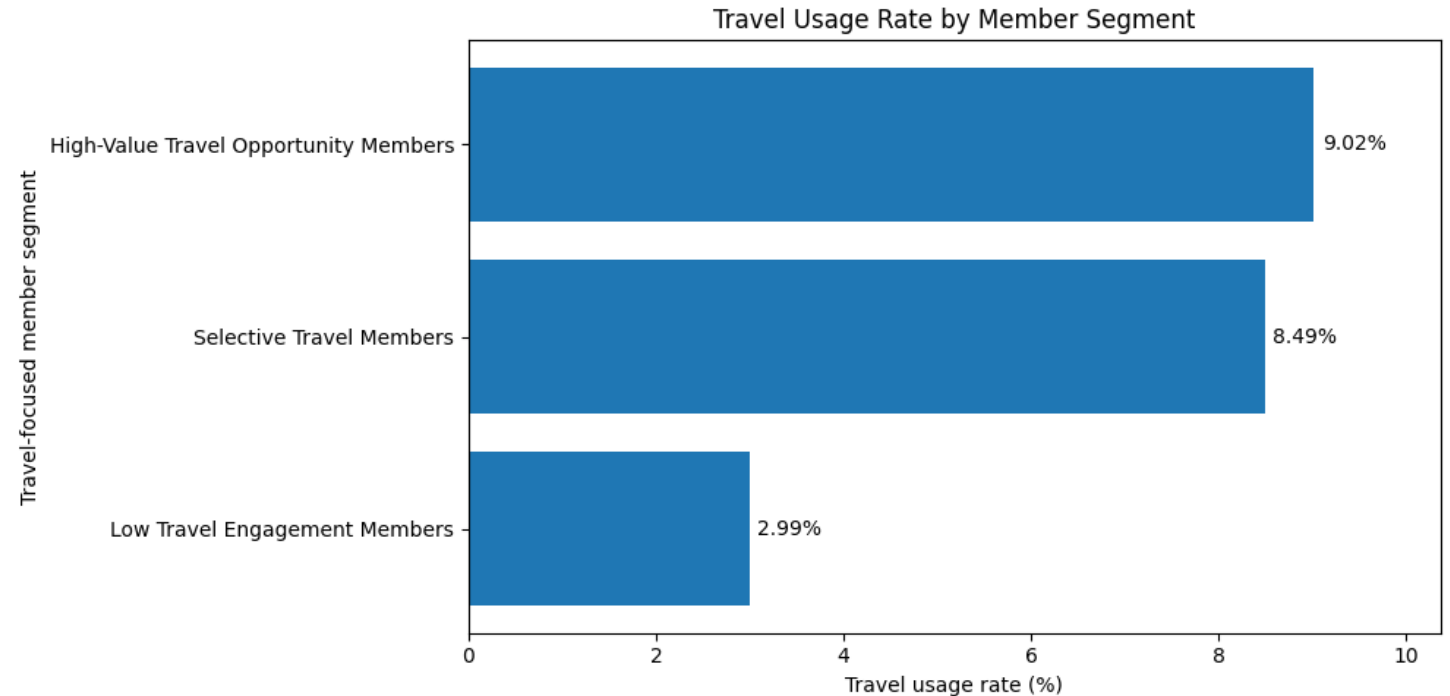
Random Forest Model Interpretation

- Member tenure is the strongest feature in the tuned Random Forest
- Age, gender, income, home ownership, and marital status add demographic context
- Branch and insurance activity provide additional service behavior signals
- The feature mix is easier to explain after excluding member transaction variables



Travel-Focused Member Segments

- Cluster analysis identified three travel-related member segments
- High-Value Travel Opportunity Members have the highest travel usage rate
- Selective Travel Members show meaningful but smaller travel engagement
- Low Travel Engagement Members are the largest group and may need awareness-focused outreach



Recommendations to AAA

- Use roadside prediction as a first service-propensity model to identify members likely to need support
- Use travel clusters to help the AAA Travel Department prioritize outreach and offers
- Target high-value travel opportunity members with stronger travel messaging
- Use lower-engagement segments for awareness and education campaigns
- Review member transaction variables separately before adding them to a production model

Limitations and Next Steps

Limitations

- Used only active members and not members who left which doesn't capture full picture
- Predicted past usage of roadside assistance, but not whether someone will need it going forward
- Longer members have simply had more time to use roadside, so tenure's strength reflects time as a member as well as true behavior
- Recall was still quite low even for best model meaning model can catch who is most likely to use roadside assistance, but not as good at catching everyone who does

Next Steps

- Include past members and test additional models for stronger results
- Reframe the target around future behavior using time-bounded target rather than past activity
- Adjust for how long members have been enrolled when measuring usage
- Include additional model types like gradient boosting to see if results improve

Conclusion

- A member-level feature table supports both classification and segmentation and allowed us predict roadside assistance usage and segment members in a useful way
- Roadside assistance usage can be predicted better than random using member-level signals
- Logistic Regression performed best overall by AUC, Accuracy, Recall, and F1 Score
- Travel clusters provide a practical segmentation path for AAA Travel Department recommendations and a good baseline for possible future service predictions

Thank You!

Questions?